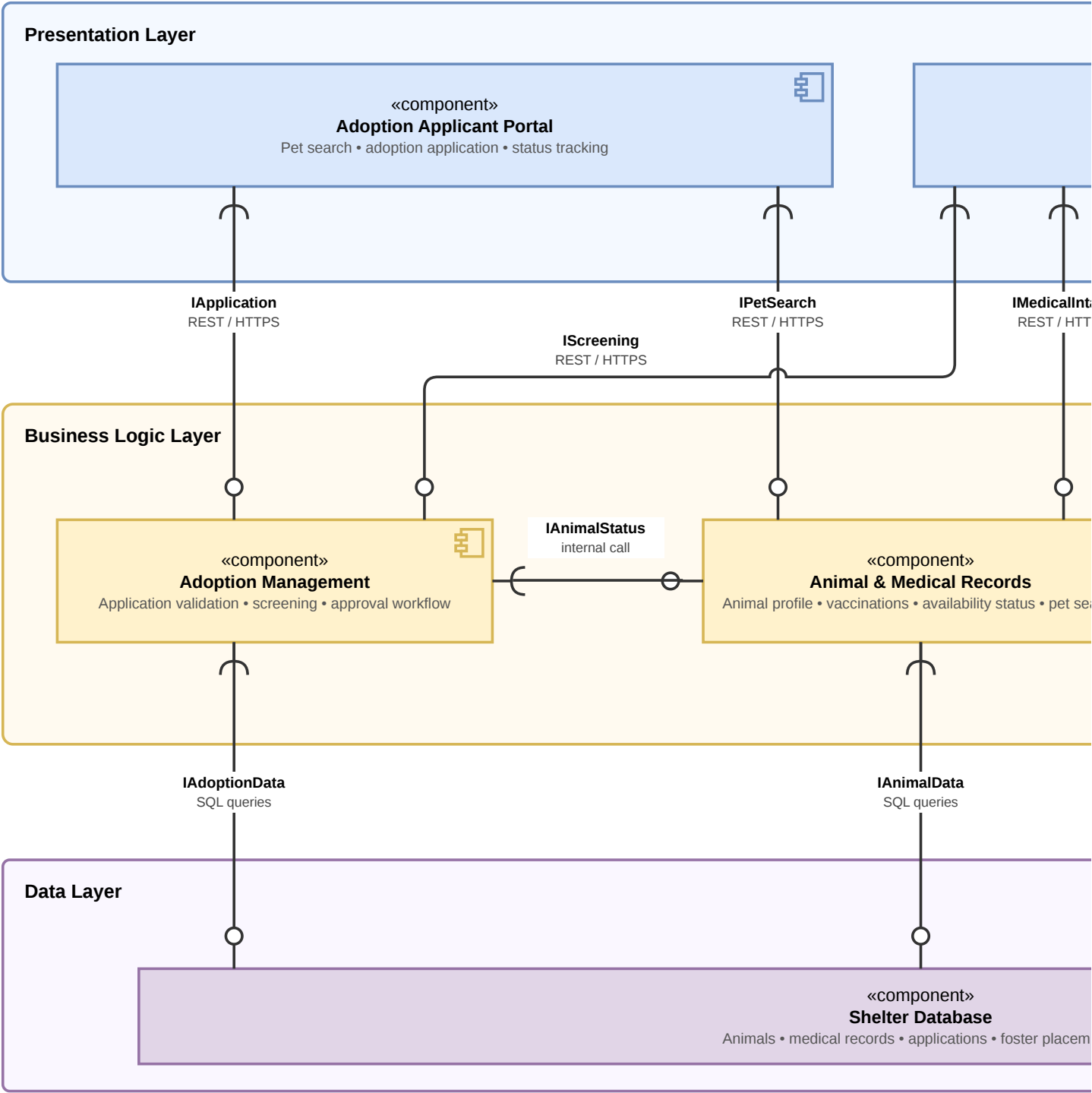


Pet Adoption & Shelter Management

UML Component Diagram — Layered Architecture

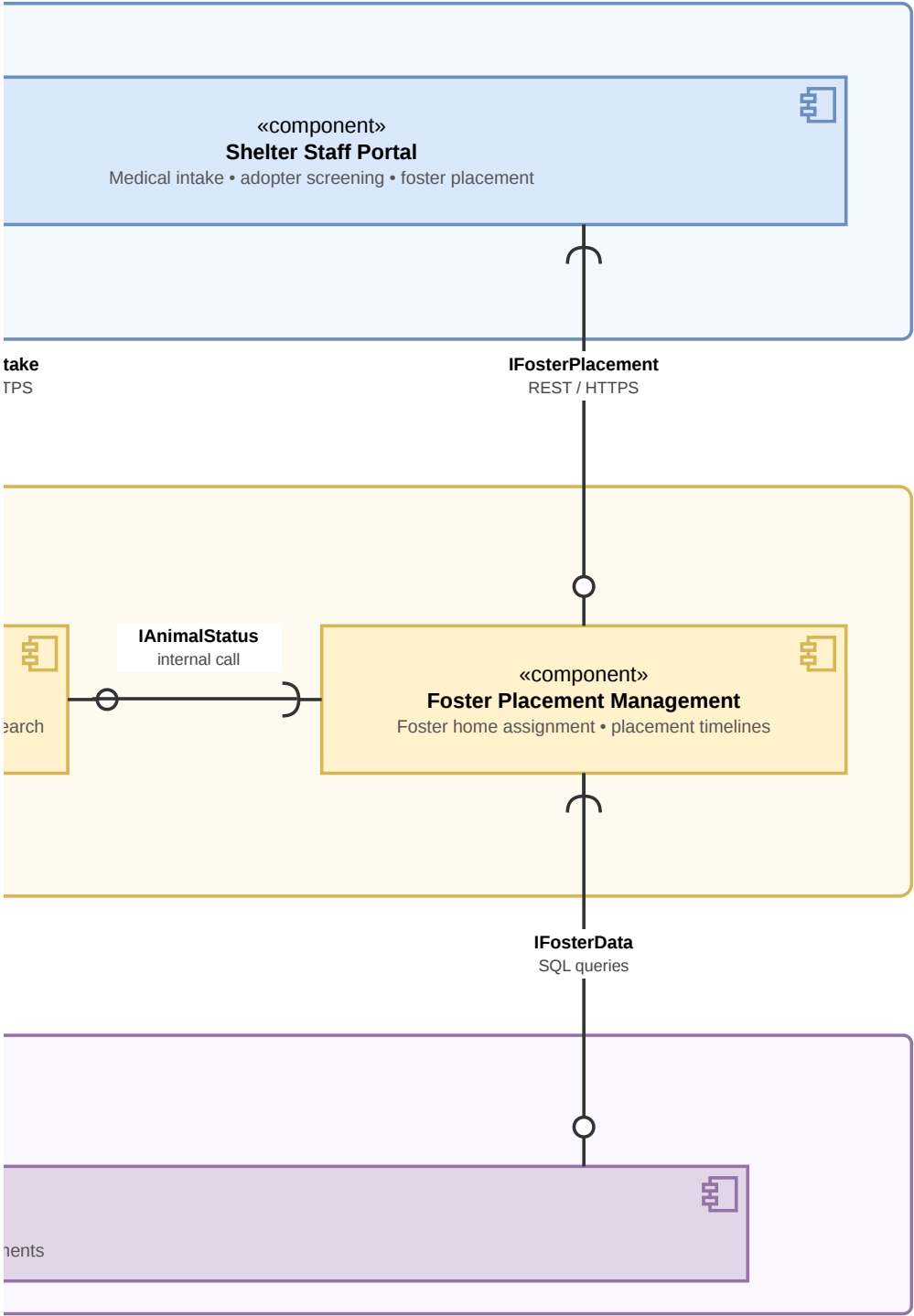


Interface catalogue (data flow: request → response)

Interface	Provided by (ball)	Required by (socket)	Protocol / tech	Data exchanged
IApplication	Adoption Management	Applicant Portal	REST / HTTPS (JSON)	adoption application + screening questionnaire
IPetSearch	Animal & Medical Records	Applicant Portal	REST / HTTPS (JSON)	filters (species, age, breed, good with kids) → ms
IScreening	Adoption Management	Staff Portal	REST / HTTPS (JSON)	questionnaire review, approve / reject → decision
IMedicalIntake	Animal & Medical Records	Staff Portal	REST / HTTPS (JSON)	intake + vaccination record → record ID, availability
IFosterPlacement	Foster Placement Mgmt	Staff Portal	REST / HTTPS (JSON)	foster home assignment, timeline dates → placement
IAAnimalStatus	Animal & Medical Records	Adoption Mgmt, Foster Placement Mgmt	internal service call	animal ID → availability / quarantine status; n
IAdoptionData	Shelter Database	Adoption Management	SQL queries	application and screening rows (create / read
IAnimalData	Shelter Database	Animal & Medical Records	SQL queries	animal profile, medical and vaccination rows
IFosterData	Shelter Database	Foster Placement Mgmt	SQL queries	foster home and placement timeline rows

ient System

re (Lab 3)



ire → application ID, status
→ paged pet list (target < 200
ision, updated status
ilability status
lacement record
mark Adopted or In Foster
d / update)

Notation

- Provided interface (ball)
- Required interface (socket)
- Assembly connector (labelled with protocol)
- «component» UML component (icon at top-right)

Layering rule: each layer only calls the layer directly below it. The IScreening connector jumps over IPetSearch where the two lines cross.

